



Wadhvani Ignite

CONCEPT BOOKLET

Module 5

Lesson 1: Overview of Prototypes & MVP

Context:

Having understood your competition and identified a potential solution, it's now time to explore prototyping and Minimum Viable Products (MVPs).

Prototyping involves creating an early model of your product to visualize and test its core features and functionalities. This early version helps bring your concept to life, allowing you to refine it based on initial feedback. An MVP, meanwhile, is the most basic form of your product that still provides value to early users and collects critical feedback. This stage is vital for confirming your assumptions, enhancing your product, and ensuring it meets user needs before moving on to full-scale development.

Program overview, and your **Venture Journey** over 14 weeks

1. Problem Identification	2. Customer Identification & Needs Validation	3. Solution Idea Generation
Week 1-2	Weeks 2-4	Week 4
■ Orientation ■ Develop Macro Industry-Problem view ■ Assemble “Real-World” Problems ■ Examine Chosen Problems ■ Analyze Problems	■ Identify customer segments ■ Assess ‘jobs-to-be-done’ ■ Develop Customer Personas ■ Understand Customer Validation ■ Validate Customer-Problem fit ■ Pivot/Refine Customer-Problem Fit for Feasibility	■ Understand Ideation techniques ■ Generate Solution Idea
4. Opportunity & Competition Mapping	5. Prototype Development & MVP	6. Opportunity Feasibility Assessment & Sizing
Week 5	Week 6	Week 7-8
■ Identify Global competitors ■ Review Industry Trends and Estimate Market Size (TAM) ■ Introduction to Financial Planning	■ Overview of Prototypes & MVP ■ Build an Initial Prototype ■ Prototype for Early Validation	■ Map your Relative Position in the Market ■ Estimate Opportunity Size
7. Business Modelling	8. Marketing & Sales Strategy	9. Financial Management for Profitability
Week 8-9	Week 9-10	Week 11-12
■ Examine Revenue Model ■ Revenue Models continued ■ Review and Organize the Lean Canvas ■ Build the LC for your Startup	■ Understand Marketing & Sales ■ Recommend a Go-To-Market (GTM) Approach ■ Map a Sales Process	■ Deep Dive into Financial Planning ■ Analyze the Path to Profitability ■ Understand Bootstrapping and Funding Options
10. Team & Talent Requirement	11. Orientation for Scale	12. Venture Idea Feasibility Presentation
Week 12	Week 13	Week 14
■ Finding Co-founders and Mentors ■ Building an Initial Team	■ Thinking Beyond the Prototype - Scale Opportunity & Orientation ■ Persuasive Story-Telling for a Viable Venture Idea ■ Pitch preparation	■ Practice Pitch ■ Final Pitches

Concepts Covered in this Lesson:

Prototyping:

- What is prototyping?
- Pros and Cons of Prototyping
- Types of Prototypes
- Understanding Fidelity

MVP

- What is MVP?
- Key features and benefits of MVP
- Key differences and similarities between Prototypes and MVP

After completing this lesson, you will be able to:

1. Understand the features and benefits of Prototypes and MVPs
2. Learn to take an iterative and user-centred approach to solution development.
3. Begin to build a Prototype of your solution idea.

Steps for you to take:

1. Review your solution idea.
2. Decide on the type of prototype you will build

What is Prototyping?

Let's dive into understanding prototypes by starting with a simple example: think of the blueprint of a building. Before construction

begins, architects create blueprints to visualize the structure, layout, and design. These blueprints allow stakeholders to see the plan, make adjustments, and address potential issues before any actual building starts.

Similarly, in product development, a prototype is a preliminary model of your product. It helps you and your team visualize and test the key features and functionalities

Corroboration is a process of confirming or verifying findings, hypotheses, or data through multiple sources or methods.

Prototypes are not just limited to product development; they are also incredibly useful in service development. Whether you are creating a physical product, a digital application, or a service offering, prototyping helps you explore ideas, test assumptions, and refine your concept.

Pros and Cons of Prototyping

Pros:

- Easy detection of missing functionality
- Getting clarity in usability/look and feel
- Better understanding of customer needs
- Aligning the team
- Reducing re-work

Cons:

- Takes time.
- Can be expensive.

Extremities are the outer limits or extreme ends of a spectrum or range, often used to identify outliers or unusual patterns.

What is fidelity in Prototyping and MVP?

Fidelity refers to the level of detail and functionality included in a prototype or an MVP. It can range from low fidelity, which is basic and simple, to high fidelity, which is more detailed and closely resembles the final product.

1. Low-Fidelity Prototypes

Low-fidelity prototypes are basic and simple representations of a product, often created using sketches, paper, or basic digital wireframes. They focus on conveying the core ideas and structure without delving into detailed design or functionality. These prototypes are quick and inexpensive to produce, making them ideal for early-stage brainstorming, concept exploration, and initial user testing. The main advantage is their flexibility, allowing for easy and rapid modifications based on feedback.

2. High-Fidelity Prototypes

High-fidelity prototypes are detailed, interactive models that closely mimic the final product in terms of appearance and functionality. They are typically created using advanced digital tools, providing a realistic user experience. High-fidelity prototypes are used for thorough testing, validating user interactions, and presenting to stakeholders. While they offer a better sense of the final product, they are more time-consuming and costly to develop, and changes based on feedback can be more complex to implement.

3. Low-Fidelity MVPs

Low-fidelity MVPs are the most basic versions of a product that include only essential features necessary to solve the primary problem for users. These MVPs are developed quickly and with minimal resources, allowing for early market testing

and feedback collection. The main benefit is the ability to validate the core concept and make informed decisions about future development without significant investment. However, they may provide a limited user experience and lack comprehensive features.

4. High-Fidelity MVPs

High-fidelity MVPs offer a more complete and polished version of the product, including more advanced features and a closer resemblance to the final product. These MVPs aim to deliver a fuller user experience and attract early adopters. They are useful for detailed market validation and can help secure investment or stakeholder buy-in. However, they require more time and resources to develop and carry a higher risk if the market response is not as expected.

What are the various types of Prototypes?

1. Paper Prototype

A paper prototype is a low-fidelity, hand-drawn version of a user interface or product. It allows designers to sketch out screens, features, and workflows on paper. This method is quick, cost-effective, and ideal for early-stage brainstorming and testing basic concepts with users.

2. Wireframe

A wireframe is a skeletal outline of a digital product, such as a website or app. It focuses on layout and functionality rather than design aesthetics. Wireframes help in organizing content and features, ensuring a clear structure before moving on to detailed design and development.

3. Device Mockup

A device mockup is a high-fidelity, often interactive, representation of a product. It looks and behaves like the final product, providing a realistic preview. Device mockups are used to demonstrate the appearance and user experience, helping stakeholders visualize the end result and identify any design issues.

4. Role Plays and Simulation

Role plays and simulations involve acting out scenarios to test and refine a service or process. Participants assume different roles to simulate real-life interactions and workflows. This method helps identify potential problems, improve service delivery, and enhance the overall user experience by seeing how the service operates in a controlled, but realistic setting.

5. Storyboards/Scenarios

Storyboards and scenarios use visual narratives to depict how users will interact with a product or service over time. They illustrate key moments and touchpoints, providing a comprehensive view of the user journey. This approach helps in understanding user needs, aligning team members, and communicating the vision clearly to stakeholders.

A pivot refers to a strategic change in a business model, product, or approach in response to market feedback or changing circumstances, aimed at improving its chances of success.

Common Low-Fidelity Prototypes

SKETCHES	PAPER	LEGO
WHY? • <i>Quickly create physical prototypes</i> • <i>Versatile, easy to modify & dismantle</i> • <i>Encourage experimentation and fun</i>	WHY? • <i>Cheap and easy to create and modify</i> • <i>Easy to get feedback on as the prototype looks rough and users don't fear hurting your feelings.</i>	WHY? • <i>Quickly create physical prototypes</i> • <i>Versatile, easy to modify & dismantle</i> • <i>Encourage experimentation and fun</i>
When? Novel Solutions	When? New digital or physical product Solution ideas	When? Novel Solutions
How? Pen and paper Sketchup Smartdraw	How? Pen and paper	How? LEGO bricks Cardboard mockups 3D printing - FreeCAD
Output: Sketch Mockup	Output: Sketch Mock-ups	Output: 3 D object (product)

STORYBOARD	DIGITAL	BLUEPRINTS
WHY? Visual plan through a combination of staging, role play and blueprint a thorough outline of the user and other stakeholder experiences	WHY? - Quickly create physical prototypes - Versatile, easy to modify & dismantle - Encourage experimentation and fun	WHY? Extensive blueprint to simulate through a combination of staging, role play and blueprint a thorough outline of the user and other stakeholder experiences
When? User Journey	When? Novel Solutions	When? Solution design for services
How? Sketch sequences. Sketchup	How? Figma Miro...	How? Sketch up Smartdraw, Planner 5D...
Output: Visual plan of user experiences/journey	Output: Wireframe / Mock-ups including clickable	Output: Blueprint of the services workflow, user experience, backstage processes...

What is MVP?

A Minimum Viable Product (MVP) is the simplest version of a product that can still deliver core value to early users. It includes just enough features to satisfy early adopters and provide feedback for future development. The primary goal of an MVP is to validate assumptions about the product's market potential and functionality with minimal resources and risk.

Key Aspects of an MVP:

- Core Functionality:** Focuses on the essential features that solve the primary problem for users. It avoids unnecessary complexity and additional features that can be added later.
- User Feedback:** Enables you to gather valuable feedback from real users early in the development process. This feedback helps in understanding user needs, preferences, and pain points.

- **Iterative Development:** Allows for quick iterations and improvements based on user feedback. This iterative process ensures the product evolves in the right direction and meets user expectations
- **Risk Reduction:** Minimizes the risk of investing time and money into a product that might not meet market needs. By testing the core idea early, you can pivot or adjust strategies as needed.
- **Market Validation:** Helps in validating the market demand and viability of the product. Early success with an MVP can attract investors and stakeholders, providing a foundation for further development.

Benefits of an MVP:

- **Speed to Market:** Gets the product into the hands of users quickly, allowing you to capitalize on market opportunities.
- **Cost Efficiency:** Reduces development costs by focusing on the most critical features initially.
- **Learning Opportunities:** Provides real-world data and insights, helping to make informed decisions about future enhancements and features.
- **Customer Engagement:** Engages early adopters who can become loyal users and advocates for your product.

An MVP is a strategic approach to product development that emphasizes learning and adapting, ensuring that you build a product that truly resonates with your target audience.

Examples:

1. **Red Bus:** The founders initially partnered with a few bus operators to list their schedules and seat availability on the website. This simple functionality addressed a major pain point for frequent travelers: the difficulty of booking bus tickets without physically going to a bus station.
2. **Zappos:** The founder of Zappos wanted to test the hypothesis that people would be willing to buy shoes online. Instead of building a full e-commerce platform, he took photos of shoes from local stores and posted them online. When someone ordered a pair, he would buy them from the store and ship them to the customer.

What are the Similarities and Differences between Prototype and MVP?

Similarities Between Prototypes and MVPs:

Validation Before Major Investment: Both prototypes and MVPs are used to validate business hypotheses before allocating substantial resources to full-scale development.

Support Iterative Development: They allow for iterative development, enabling adjustments based on feedback to improve the product incrementally.

Reduce Initial Costs: Both approaches aim to minimize the initial investment required, both in terms of time and money, reducing risk in the early stages of product development.

Differences Between Prototypes and MVPs:

Purpose and Testing: Prototypes are generally used internally to validate design concepts and ensure the product works technically (usually low to medium fidelity). They help the development team identify and fix design flaws before advancing to later stages.

User Interaction and Validation: An MVP, on the other hand, is a more complete version of the product with enough features to attract early adopters and validate a product-customer fit in the real market (high fidelity). It is tested with actual target users to gather feedback on its viability and to see how it meets the market needs.

Fidelity and Scope: Prototypes can range from simple sketches or paper interfaces to more detailed digital mock-ups, primarily focused on exploring ideas and refining them internally. MVPs are functional products released to the market to test a business model with real customers, focusing more broadly on early market penetration and learning.

Call-To-Action:

Complete Venture Activity 5.1- Build Your Prototype

Choose Your Prototype Type: Discuss with your team members and decide on the type of prototype you will be building. Consider factors such as the complexity of your solution, available resources, and the preferences of your team.

Gather Materials: Collect the necessary materials for building your chosen prototype type. Ensure you have everything you need to bring your prototype to life.

Start Building your Prototype: Begin the process of building your prototype. Collaborate with your team members to bring your ideas to fruition. Focus on visualizing and demonstrating key features and functionalities of the solution.

Iterate and Refine: As you build your prototype, continuously iterate and refine your design based on feedback and testing. Be open to adjusting and improving your prototype to ensure it effectively communicates the intended user experience.